

SC 5392&1
WASSCE 2026
PHYSICS 2&1
Essay and Objective
2 hours 55 minutes

2&1

Name

Index Number

THE WEST AFRICAN EXAMINATIONS COUNCIL

**West African Senior School Certificate Examination
for School Candidates**

SC 2026

PHYSICS 2&1
ESSAY AND OBJECTIVE

2 hours 55 minutes

*Do **not** open this booklet until you are told to do so. While you are waiting, read and observe the following instructions carefully. Write your **name** and **index number** in **ink** in the spaces provided above.*

*This booklet consists of **two** papers. Answer **Paper 2** which comes first, in your answer booklet and **Paper 1** on your Objective Test answer sheet. **Paper 2** will last $1\frac{1}{2}$ hours after which the answer booklet will be collected. Do **not** start **Paper 1** until you are told to do so. **Paper 1** will last $1\frac{1}{4}$ hours.*

Answer **twelve** questions in all; **nine** questions from Part A and **three** questions from Part B.

PART A
[20 marks]

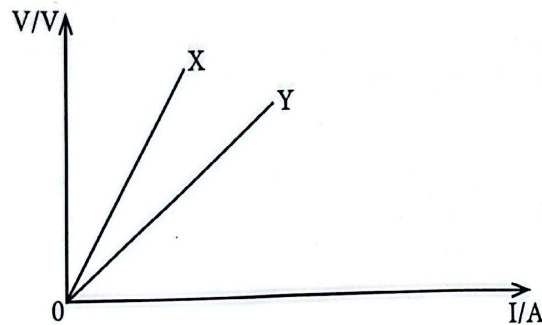
Answer **all** questions from this part.

1. What is *half-life* of a radioactive atom? [2 marks]
2. Name the charge carriers in **each** of the following materials:
(a) metals;
(b) semi-conductors. [2 marks]
3. A projectile is launched at an angle of 30° to the horizontal with an initial velocity of 40 m s^{-1} . Calculate the range of the projectile. [$g = 10 \text{ m s}^{-2}$] [2 marks]
4. State the thermometric property for **each** of the following thermometers:
(a) Constant volume gas thermometer;
(b) Resistance thermometer. [2 marks]
5. A current carrying conductor of length 2 m placed perpendicular to a uniform magnetic field of flux density 2 T experiences a force of 20 N. Calculate the current flowing through the conductor. [2 marks]
6. List **two** applications of echo. [2 marks]
7. A spring is stretched 3 times its original length. Calculate the strain in the spring. [2 marks]
8. A metal has a work function of 2.0 eV. An ultraviolet light of photon energy 4.5 eV shines on the surface of the metal. Calculate the maximum energy of the emitted photoelectrons. [$e = 1.6 \times 10^{-19} \text{ C}$] [3 marks]
9. A progressive wave travels 25 cm in 0.5 seconds. Calculate the frequency of the wave if the distance between two successive troughs is 0.4 cm. [3 marks]

Answer three questions from this part.
All questions carry equal marks.

10. (a) A rectangular sheet of metal 25.0 cm long and 20.0 cm wide initially at a temperature of 35 °C is heated to a final temperature of 120 °C. Calculate the increase in area of the sheet of metal. [$\alpha = 1.5 \times 10^{-5} \text{ K}^{-1}$] [3 marks]
- (b) A variable alternating signal $V = 25 \sin 100\pi t$ is applied to a purely capacitive circuit. If the peak current in the circuit is 14.0 mA, calculate the capacitance of the capacitor. [$\pi = 3.14$] [3 marks]
- (c) (i) State two pieces of information that could be obtained from a velocity-time graph.
(ii) For each of the information stated in (i), show how it could be obtained. [3 marks]
- (d) Explain how radiation containing alpha, beta and gamma would be affected by an electric field. [3 marks]
- (e) The energy, E_n of a given state of principal quantum number, n of a hydrogen-like atom is given by $E_n = \frac{-122}{n^2} \text{ eV}$. Draw an energy level diagram to show the transition of an electron from the ground state to the second excited state. [4 marks]
- (f) A fisherman counts 180 sea waves in 1 minute. The distance between four successive crests is 12 metres. Calculate the speed of the wave. [4 marks]
11. (a) A mercury column of height, h exerts a pressure of $3.4 \times 10^4 \text{ Pa}$ at the base of a vessel. Calculate the value of h . [Density of mercury = $13.6 \times 10^3 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$] [3 marks]
- (b) A temperature of 50 °C produces a fractional change in volume of 7.5×10^{-3} for a given metal sphere. Calculate the linear expansivity of the material of the sphere. [3 marks]
- (c) An electrical source generates 50 kW of power at a voltage of 10 kV. The power is fed through a transmission cable of resistance 2 Ω . Calculate the power loss in the cable. [3 marks]
- (d) An x-ray tube is operated by the application of 25 kV across the tube. Calculate the shortest wavelength of photon radiated. [$c = 3.0 \times 10^8 \text{ m s}^{-1}$, $e = 1.6 \times 10^{-19} \text{ C}$, $h = 6.6 \times 10^{-34} \text{ J s}$] [3 marks]
- (e) (i) State the condition necessary for a transformer to be 100 % efficient.
(ii) For an ideal transformer in operation, show that the turns ratio of the coil is related to the ratio of input current, I_1 to the output current, I_2 . [4 marks]
- (f) A man walks 4 km due east and then walks 5 km due south. He further walks x km due east such that the magnitude of his displacement is 8.6 km.
(i) Sketch a vector diagram for the motion.
(ii) Determine the value of x . [4 marks]

12. (a) Two electrically charged parallel plates are 1.5 cm apart in vacuum. If an electron falls between the plates which are maintained at 20 V, calculate the magnitude of the force on the electron. [$e = 1.6 \times 10^{-19} \text{ C}$] [3 marks]
- (b) A plane progressive wave is represented by the equation $y = 0.2 \sin(200\pi t + \frac{2}{5}\pi x)$, where y and x are in centimetres and t in seconds.
 (i) Determine the velocity of the wave.
 (ii) Sketch the displacement graph of the wave, indicating the amplitude and the wavelength for one cycle. [4 marks]
- (c) The Bohr energy of the hydrogen atom is given by $E_n = \frac{-13.6}{n^2} \text{ eV}$. Given that the hydrogen atom undergoes de-excitation from the second excited state, calculate the frequency of the photon released. [$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$, $h = 6.6 \times 10^{-34} \text{ J s}$] [3 marks]
- (d) The rise in the body temperature of a baby over the normal temperature of 37°C is 3.6°F . Calculate the final temperature of the baby in degree celsius. [3 marks]
- (e) The graph below shows a variation of voltage, V with current, I for two materials, X and Y.



- (i) State **two** physical quantities that could be deduced from the graph.
 (ii) Which of the materials would be **more** suitable for making:
 (α) connecting wires;
 (β) standard resistors. [4 marks]
- (f) State **three** applications of friction in daily human activities. [3 marks]
13. (a) (i) State *Newton's second law of motion*.
 (ii) Use the law stated in (i) to show the principle of conservation of momentum. [4 marks]
- (b) Explain **briefly** why a capacitor prevents a direct current but allows an alternating current to pass through it. [3 marks]

- (c) A piece of ancient bone from a site showed a carbon-14 activity of 9.5 decays per minute per gram. A bone specimen from a living animal showed a carbon-14 activity of 12.0 decays per minute per gram. Estimate the age of the ancient bone.
[Half-life of carbon-14 = 5570 years] [4 marks]
- (d) Two point charges +32 nC and +20 nC are separated by a distance of 0.6 m in vacuum. Determine the position of the neutral point from the +32 nC charge. [$k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$] [3 marks]
- (e) An electromagnetic radiation is incident on three different photosensitive surfaces, A, B and C.
Use Einstein's photoelectric equation to explain each of the following observations:
(i) photoelectrons emitted from A have zero kinetic energy;
(ii) no photoelectrons were emitted from B;
(iii) photoelectrons emitted from C have some kinetic energy. [3 marks]
- (f) (i) State the *Principle of moments*.
(ii) A uniform bar of length 4.0 m is pivoted at its centre. A 30.0 N weight hangs 1.0 m left of the pivot. How far must a 50.0 N weight be hung to balance the bar horizontally? [3 marks]

END OF ESSAY TEST

**DO NOT TURN OVER THIS PAGE
UNTIL YOU ARE TOLD TO DO SO.**

**YOU WILL BE PENALIZED SEVERELY IF YOU ARE
FOUND LOOKING AT THE NEXT PAGE BEFORE
YOU ARE TOLD TO DO SO.**

**WHILE YOU ARE WAITING, READ THE FOLLOWING
INSTRUCTIONS CAREFULLY.**

PAPER 1
OBJECTIVE TEST
[50 marks]

1¼ hours

1. Use **2B** pencil throughout.
2. On the pre-printed answer sheet, check that the following details are **correctly** printed:
 - (a) In the space marked **Name**, check your **surname** followed by your **other names**.
 - (b) In the spaces marked *Examination*, *Year*, *Subject* and *Paper*, check 'WASSCE (SC)', '2026', 'PHYSICS', and '1' respectively.
 - (c) In the box marked *Index Number*, your **index number** has been printed vertically in the spaces on the left-hand side, and each numbered space has been shaded in line with each digit. **Reshade** each of the shaded spaces.
 - (d) In the box marked *Subject Code*, the digits 539113 are printed vertically in the spaces on the left-hand side. **Reshade** the corresponding numbered spaces as you did for your index number.
3. An example is given below. This is for a male candidate whose name is Tony Mensah ATIAH. His *index number* is 7102143958 and he is offering *Physics 1*.

THE WEST AFRICAN EXAMINATIONS COUNCIL
ANSWER SHEET

PRINT IN BLOCK LETTERS	
Name: ATIAH TONY MENSAH	GHA
Examination: WASSCE (SC)	Year: 2026
Subject: PHYSICS	Paper: 1

INSTRUCTIONS TO CANDIDATES

1. Use grade 2B pencil throughout.
2. Answer each question by choosing one letter and shading like this: ☐ A ☒ B ☐ C ☐ D
3. Erase completely any answer you wish to change.
4. Leave extra spaces blank if the answer spaces provided are more than you need.
5. Do not make any markings across the heavy black marks at the right hand edge of your answer sheet.

INDEX NUMBER										SUBJECT CODE											
7	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9	5	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9
1	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9	3	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9
0	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9	9	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9
2	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9	1	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9
1	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9	1	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9
4	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9	3	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9
3	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9											
9	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9											
5	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9											
8	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9											

For Supervisors only

If candidate is absent shade this space ☐

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Answer all the questions.

Each question is followed by four options lettered A to D. Find the correct option for each question and shade in pencil on your answer sheet the answer space which bears the same letter as the option you have chosen.

Give only one answer to each question. An example is given below.

A body starts moving with a speed of 40 m s^{-1} and accelerates uniformly to 90 m s^{-1} in 4.0 s .

Calculate the distance travelled.

- A. 260 m
- B. 260 m
- C. 200 m
- D. 180 m

The correct answer is 260 m, which is lettered B, and therefore answer space B would be shaded.

☐ ☒ ☐ ☐

Think carefully before you shade the answer spaces; erase completely any answers you wish to change.

Do all rough work in this question paper.

Now answer the following questions:

1. The major limitation to the Rutherford's model of the atom is that, he could not explain why the atom
 - A. occupies a small volume though charged.
 - B. has its electrons surrounding the nucleus.
 - C. only emits light of specific frequency.
 - D. has its entire mass concentrated at the centre.
2. The SI unit for measuring the hotness of a body is
 - A. Celsius.
 - B. degree.
 - C. kelvin.
 - D. fahrenheit.
3. Kirchhoff's junction rule is a direct application of the principle of conservation of
 - A. energy.
 - B. electromotive force.
 - C. charges.
 - D. current.
4. Which atomic model introduced the idea of electrons orbiting the nucleus in fixed energy levels?
 - A. Bohr model
 - B. Dalton model
 - C. Thompson model
 - D. Rutherford model
5. A body continues to sink in a fluid until it displaces its own weight of the fluid. This is a statement of the principle of
 - A. Archimedes.
 - B. Density.
 - C. Flotation.
 - D. Upthrust.

6. Under a constant tension, a plucked wire produces a note 300 Hz when the length is 2 m. At what length is the frequency 400 Hz?
- 1.3 m
 - 1.5 m
 - 2.7 m
 - 3.0 m
7. A light ray passes from air into a medium at an angle of 30° . The angle of refraction in the medium is 20° . Calculate the refractive index of the medium.
- 1.15
 - 1.46
 - 1.52
 - 1.74
8. Which of the following instruments could generate parallax error when in use?
- Protractor
 - Electronic balance
 - Metre rule
 - Pair of vernier calipers
- I and III only
 - I and IV only
 - I, III and IV only
 - I, II and IV only
9. Which of the following statements about a laser beam is **not** correct?
- It is a parallel coherent beam of high intensity
 - It is produced by a process of stimulated emissions of radiation
 - Its intensity is practically unchanged with distance
 - It is a device which produces a collimated monochromatic light
10. The nucleus of an atom remains stable despite the repulsion among protons because the
- neutrons cancel out the charges of the protons.
 - electrons surrounding the nucleus hold it together.
 - strong nuclear force binds protons and neutrons tightly.
 - protons lose their charge when packed closely.
11. The condition under which Kirchoff's mesh rule holds is that, there must be a
- junction in the network.
 - source of emf.
 - closed loop in the circuit.
 - load in the circuit.
12. Water of mass 250 g at 20°C is thoroughly mixed with water of mass 200 g at 50°C until thermal equilibrium is established at a temperature $\theta^\circ\text{C}$. Assuming no heat was lost to the surrounding, calculate the value of $\theta^\circ\text{C}$.
- 25.0°C
 - 30.0°C
 - 33.3°C
 - 35.0°C

13. A radioactive substance has a half-life of 4 years. If the initial amount of the substance is 160 g, how much of the substance would remain after 12 years?
- 20 g
 - 40 g
 - 60 g
 - 80 g
14. A particle with charge 1.6×10^{-19} C enters a uniform magnetic field of magnitude 0.2 T at a speed of 4.0×10^6 m s⁻¹. If the angle between the particle's velocity and the magnetic field is 30°, determine the magnitude of the force acting on the particle.
- 1.6×10^{-3} N
 - 5.0×10^{-14} N
 - 6.4×10^{-14} N
 - 8.0×10^{-14} N
15. The state of matter for which the intermolecular attraction is weak with moderate intermolecular separation is
- gas.
 - liquid.
 - solid.
 - plasma.
16. A scientist observes a substance filling its container evenly but compressible under pressure. Which of the statements below **correctly** links this behaviour to its molecular arrangement?
- Molecules are tightly packed in fixed positions.
 - Molecules are far apart with weak forces.
 - Molecules pass each other due to strong forces.
 - Molecules form strong bonds about their lattice.
17. A transformer operates at 50 Hz. If the frequency is increased to 60 Hz while keeping voltage constant, how do the core losses change? It
- increases.
 - decreases.
 - remains the same.
 - depends on the load current.
18. A piece of metal brought near a suspended bar magnet shows a strong attraction. It can be concluded that the piece of metal is a
- magnetic material.
 - magnet with same polarity.
 - magnet with opposite polarity.
- I only
 - II only
 - I and II only
 - I and III only
19. An electron transitions from $n = 4$ to $n = 2$. Calculate the energy of the photon emitted.
- $$[E_n = \frac{-13.6}{n^2} \text{ eV, } 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}]$$
- 2.55×10^{-19} J
 - -6.0×10^{-19} J
 - 4.09×10^{-19} J
 - -2.55×10^{-19} J



- a. None; none of the three authors has been convicted of a crime.
 b. None; none of the three authors has been convicted of a crime.
 c. None; none of the three authors has been convicted of a crime.
 d. None; none of the three authors has been convicted of a crime.

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1. The first step is to identify the problem.
2. The second step is to define the problem.
3. The third step is to analyze the problem.
4. The fourth step is to develop a solution.

What is the average number of people who are in the same situation as you?

- A. *Inter-uterine*
 B. *P-T, posterior-lateral*
 C. *P-T-T, posterior*
 D. *Anterior-lateral*

1

The following findings were made by the FBI in its investigation of the above-captioned matter. It is noted that the Bureau has received information from the FBI in New York City that the above-captioned matter is being handled by the New York City Office.

-

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3

WILSON

- A. ~~When the company's revenue is low, the price of the product is high.~~
 B. ~~When the company's revenue is low, the price of the product is low.~~
 C. ~~When the company's revenue is high, the price of the product is high.~~
 D. ~~When the company's revenue is high, the price of the product is low.~~

[illegible]

- | | |
|---|------|
| A | 50% |
| B | 70% |
| C | 50% |
| D | 100% |

2

Figure 1. Schematic representation of the experimental design. The subjects were divided into two groups: the control group (CG) and the experimental group (EG). The CG was exposed to a control environment (CE) and the EG was exposed to an experimental environment (EE). The EE was designed to simulate a real-world environment with various stimuli (e.g., visual, auditory, and olfactory) and tasks (e.g., navigation, decision-making, and communication). The subjects were exposed to the EE for a period of 24 hours. The results of the experiment are shown in the bar chart, which displays the mean values of the various parameters measured for each group. The parameters measured include heart rate, blood pressure, respiratory rate, and oxygen saturation. The EG showed significantly higher values for all parameters compared to the CG, indicating a physiological response to the EE.

-

- | | |
|---|--------------|
| A | Normal Heart |
| B | Normal Heart |
| C | Normal Heart |
| D | Normal Heart |

28. Two identical charges are placed close to each other in an electric field. Which of the following statements about the charges are correct?
- There exists a neutral point between them
 - They attract each other with an equal force
 - They repel each other with the same amount of force
 - The mutual force between them obeys the inverse square law
- I, II and IV only
 - I, III and IV only
 - I and III only
 - I and IV only
29. A car's speedometer shows that the car is moving at a speed of 7200 millimetres per second. This speed in kilometres per hour is
- 7.2 km h⁻¹.
 - 72.0 km h⁻¹.
 - 43.2 km h⁻¹.
 - 25.9 km h⁻¹.
30. When a charged body touches the metal case of a neutral electroscope, the gold leaf is
- charged.
 - uncharged.
 - diverged.
 - collapsed.
31. Which of the following pairs of quantities is dimensionally equivalent to Young's modulus of elasticity?
- Pressure and tensile stress
 - Tensile stress and power
 - Pressure and tensile strain
 - Energy and tensile strain
32. A galvanometer (0–5 mA) with a resistance of 2 Ω is to function as (0–10 V) instrument. Which of the following conditions must be satisfied for the purpose?
- A 1998 Ω resistor is connected in series with the galvanometer
 - A 1998 Ω resistor is connected across the galvanometer
 - A 2 Ω standard resistor is connected in series with the galvanometer
 - A 2 Ω standard resistor is connected across the galvanometer
33. Which of the statements below best explains how electric charges are distributed on different conductor shapes?
- Charges accumulate evenly on all surfaces regardless of shape
 - Charges are more concentrated on flatter surfaces
 - Charges distribute uniformly on curved non-metallic surfaces
 - Charges tend to concentrate more at sharp points
34. Two plane mirrors inclined at an angle θ to each other eventually produce 12 images. The value of θ is
- 27.62°
 - 27.69°
 - 29.91°
 - 32.72°
35. A point charge is placed in an electric field. What is the electric field intensity at a point 0.2 m away from the point charge of magnitude 3.0×10^{-6} C? [$(4\pi\epsilon_0)^{-1} = 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$]
- $3.00 \times 10^{-5} \text{ N C}^{-1}$
 - $4.50 \times 10^5 \text{ N C}^{-1}$
 - $6.75 \times 10^{-5} \text{ N C}^{-1}$
 - $6.75 \times 10^5 \text{ N C}^{-1}$

36. Projectile motion is based on the assumption that, the only force acting on the body in the motion is **solely**
- the body's weight.
 - the viscous drag.
 - air resistance.
 - acceleration.
37. A solid iron ball weighs 50 N in air and 40 N when fully submerged in water. What is the buoyant force acting on the ball?
- 10 N
 - 20 N
 - 30 N
 - 40 N
38. All the following statements are true about magnetic field lines **except**
- they do not cross each other.
 - they form close loops from the north pole to the south pole.
 - they form close loops from the south pole to the north pole.
 - the flux density shows the strength of the magnetic field.
39. A technician wants to store sound recordings with minimal noise interference. Which type of signal is more suitable and why?
- Analogue signal because it has smooth and continuous variation.
 - Analogue signal because it resists interference better.
 - Digital signal because its discrete values are less affected by noise.
 - Digital signal because it uses discrete values that are less affected by noise.
40. Which of the following statements explains why LEDs are preferred in electronic displays?
- LEDs are costlier but emit more heat
 - LEDs glow with low energy
 - LEDs use more power
 - LEDs work only in reverse bias mode
41. Which of the following physical quantities is basic?
- Force
 - Energy
 - Power
 - Mass
42. Five identical capacitors are connected in parallel to a source in a circuit. If the effective capacitance is $200\ \mu\text{F}$, calculate the capacitance of **each** of the capacitors.
- $5.0 \times 10^{-9}\ \text{F}$
 - $2.5 \times 10^{-8}\ \text{F}$
 - $4.0 \times 10^{-5}\ \text{F}$
 - $2.0 \times 10^{-4}\ \text{F}$
43. In which of the following materials could the charge carriers be controlled by temperature?
- Conductor
 - Semiconductor
 - Insulator
 - Superconductor
44. What is meant by *the moment of force*?
- The ratio of force to the perpendicular distance from the pivot
 - The product of force and the perpendicular distance from the pivot
 - The force required to produce a turning effect
 - The amount of energy an object has

45. When a transistor is forward-biased at the base-emitter junction and reverse-biased at the collector-base junction, it
- A. enters cut off mode.
 - B. amplifies signals.
 - C. acts as a resistor.
 - D. switches off.
46. Which of the situations below demonstrates Newton's first law of motion?
- A. Soccer ball remains still until it is kicked
 - B. A car speeds up as it moves downhill
 - C. A person lifts a box with more force to move it faster
 - D. A soccer ball kicked into the air
47. Two race cars collide on a frictionless track. After the collision, the cars stick together and move with a common velocity. Which type of collision does this scenario describe?
- A. Elastic, because kinetic energy is conserved
 - B. Inelastic, because momentum is lost
 - C. Elastic, because momentum is lost
 - D. Inelastic, because kinetic energy is lost
48. When a magnet placed near a coil is moved through the coil, it
- A. produces light.
 - B. stops current flow.
 - C. produces a current.
 - D. reduces resistance.
49. Which of the following terms is **not** a property of a longitudinal wave?
- A. Reflection
 - B. Diffraction
 - C. Polarization
 - D. Compression
50. The region around a bar magnet where the lines of force are closest is the
- A. field.
 - B. neutral point.
 - C. pole.
 - D. meridian.

END OF PAPER